Car Seat Sales Forecating Using CART Regression Tree

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Contents

Executive Summary

This project develops a **CART (Classification and Regression Tree)** regression model to forecast car seat sales across retail stores using demographic, pricing, and merchandising variables.

Using **5-fold cross-validation**, the model identifies key drivers of sales performance, including shelf location quality, pricing strategy, and local demographic factors.

The trained model is applied to **new, unseen store data** to generate forward-looking sales forecasts that can support:

- Inventory planning and demand forecasting
- Store-level pricing strategy
- Targeted advertising budget allocation

This CART model serves as an **interpretable baseline**, establishing a foundation for future comparison with advanced models such as **Random Forests and Neural Networks**.

1. Project Overview

Accurately forecasting product sales is critical for inventory planning, pricing strategy, and marketing optimization.
This project focuses on predicting **car seat sales** using store-level, demographic, and marketing-related attributes.

This notebook establishes a **baseline predictive model** using a **CART (Classification and Regression Tree)** approach.
CART is intentionally chosen as the first model due to its interpretability and ability to capture non-linear relationships.

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2. Business Problem

Retail managers need to understand: - Which factors most strongly influence car seat sales - How pricing, advertising, and demographics interact - Whether interpretable models can provide actionable insights

The goal is to: - Predict expected sales volume
- Establish a baseline benchmark for more advanced models
- Identify key drivers of sales performance

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3. Dataset Description

The dataset contains store-level information related to pricing, competition, demographics, and marketing spend.

Target Variable

- **Sales**: Number of car seats sold (measured in hundreds of units)

Predictor Variables

Variable	Description
Store	Unique store identifier (excluded from modeling)
CompPrice	Competitor's price
Income	Local average income (in $000s)
Advertising	Advertising budget (in $000s)
Population	Local population size (in $000s)
Price	Store's price
ShelfLoc	Shelf location quality (Good, Medium, Bad)
Age	Average local age
Education	Average years of education
Urban	Urban location (Yes/No)
US	Store located in US (Yes/No)

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4. Environment Setup

```r
library(caret)
library(rpart)
```

```r
library(rpart.plot)
library(dplyr)
library(ggplot2)
```

5. Data Ingestion

```r
car_seat_sales <- read.csv("data/car_seat_sales.csv")

# Preview the data
head(car_seat_sales)
```

```
##   Store CompPrice Income Advertising Population Price ShelfLoc Age Education
## 1     1       138     73          11        276   120      Bad  42        17
## 2     2       111     48          16        260    83     Good  65        10
## 3     3       113     35          10        269    80   Medium  59        12
## 4     4       117    100           4        466    97   Medium  55        14
## 5     5       141     64           3        340   128      Bad  38        13
## 6     6       124    113          13        501    72      Bad  78        16
##   Urban  US Sales
## 1   Yes Yes     9
## 2   Yes Yes    16
## 3   Yes Yes    16
## 4   Yes Yes     0
## 5   Yes  No     3
## 6    No Yes     5
```

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6. Data Preparation

- Remove non-predictive identifier variables
- Let CART internally handle categorical variables
- Preserve all relevant features for interpretability

```r
library(dplyr)

# Remove Store ID
model_data <- car_seat_sales %>%
  select(-Store)

str(model_data)
```

```
## 'data.frame':    400 obs. of  11 variables:
##  $ CompPrice  : int  138 111 113 117 141 124 115 136 132 132 ...
##  $ Income     : int  73 48 35 100 64 113 105 81 110 113 ...
##  $ Advertising: int  11 16 10 4 3 13 0 15 0 0 ...
##  $ Population : int  276 260 269 466 340 501 45 425 108 131 ...
##  $ Price      : int  120 83 80 97 128 72 108 120 124 124 ...
##  $ ShelfLoc   : chr  "Bad" "Good" "Medium" "Medium" ...
##  $ Age        : int  42 65 59 55 38 78 71 67 76 76 ...
##  $ Education  : int  17 10 12 14 13 16 15 10 10 17 ...
##  $ Urban      : chr  "Yes" "Yes" "Yes" "Yes" ...
##  $ US         : chr  "Yes" "Yes" "Yes" "Yes" ...
```

```
## $ Sales      : int  9 16 16 0 3 5 3 3 4 2 ...
```

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7. Modeling Strategy

Why CART?

CART was selected as the baseline modeling approach due to its:

- Ability to capture **non-linear relationships**
- Automatic detection of **feature interactions**
- High interpretability through decision rules
- Suitability as a benchmark before ensemble methods

Validation Approach

To ensure robust model performance, a **5-fold cross-validation** strategy was used:

- Reduces overfitting
- Provides stable performance estimates
- Ensures generalizability beyond training data

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8. Model Training – CART Regression Tree

```r
set.seed(1234)

library(caret)

cv_method <- trainControl(
  method = "cv",
  number = 5
)

cart_model <- train(
  Sales ~ .,
  data = model_data,
  method = "rpart",
  trControl = cv_method,
  tuneLength = 100
)

cart_model
```

```
## CART
##
## 400 samples
##  10 predictor
##
## No pre-processing
## Resampling: Cross-Validated (5 fold)
## Summary of sample sizes: 319, 322, 320, 319, 320
## Resampling results across tuning parameters:
##
##   cp          RMSE      Rsquared     MAE
```

```
## 0.0000000000 5.866129 0.007217117 4.825055
## 0.0001428212 5.866129 0.007217117 4.825055
## 0.0002856424 5.866129 0.007217117 4.825055
## 0.0004284635 5.866129 0.007217117 4.825055
## 0.0005712847 5.866129 0.007217117 4.825055
## 0.0007141059 5.866129 0.007217117 4.825055
## 0.0008569271 5.866129 0.007217117 4.825055
## 0.0009997483 5.866129 0.007217117 4.825055
## 0.0011425694 5.866129 0.007217117 4.825055
## 0.0012853906 5.866129 0.007217117 4.825055
## 0.0014282118 5.866129 0.007217117 4.825055
## 0.0015710330 5.866129 0.007217117 4.825055
## 0.0017138541 5.866129 0.007217117 4.825055
## 0.0018566753 5.866129 0.007217117 4.825055
## 0.0019994965 5.866129 0.007217117 4.825055
## 0.0021423177 5.866129 0.007217117 4.825055
## 0.0022851389 5.866129 0.007217117 4.825055
## 0.0024279600 5.866129 0.007217117 4.825055
## 0.0025707812 5.866129 0.007217117 4.825055
## 0.0027136024 5.866129 0.007217117 4.825055
## 0.0028564236 5.866129 0.007217117 4.825055
## 0.0029992448 5.866129 0.007217117 4.825055
## 0.0031420659 5.866129 0.007217117 4.825055
## 0.0032848871 5.866129 0.007217117 4.825055
## 0.0034277083 5.866129 0.007217117 4.825055
## 0.0035705295 5.866129 0.007217117 4.825055
## 0.0037133506 5.866129 0.007217117 4.825055
## 0.0038561718 5.866129 0.007217117 4.825055
## 0.0039989930 5.859179 0.007146496 4.816305
## 0.0041418142 5.859179 0.007146496 4.816305
## 0.0042846354 5.859179 0.007146496 4.816305
## 0.0044274565 5.859179 0.007146496 4.816305
## 0.0045702777 5.869461 0.006374499 4.821714
## 0.0047130989 5.869461 0.006374499 4.821714
## 0.0048559201 5.878026 0.006425628 4.828836
## 0.0049987413 5.878026 0.006425628 4.828836
## 0.0051415624 5.878026 0.006425628 4.828836
## 0.0052843836 5.878026 0.006425628 4.828836
## 0.0054272048 5.878026 0.006425628 4.828836
## 0.0055700260 5.880756 0.005848045 4.830845
## 0.0057128471 5.880756 0.005848045 4.830845
## 0.0058556683 5.880756 0.005848045 4.830845
## 0.0059984895 5.880756 0.005848045 4.830845
## 0.0061413107 5.880756 0.005848045 4.830845
## 0.0062841319 5.880756 0.005848045 4.830845
## 0.0064269530 5.888233 0.006183699 4.841920
## 0.0065697742 5.894653 0.006387557 4.840366
## 0.0067125954 5.881321 0.006344049 4.822340
## 0.0068554166 5.873247 0.006944273 4.816679
## 0.0069982378 5.873247 0.006944273 4.816679
## 0.0071410589 5.857917 0.006801125 4.805772
## 0.0072838801 5.857917 0.006801125 4.805772
## 0.0074267013 5.857917 0.006801125 4.805772
## 0.0075695225 5.859914 0.006607028 4.814049
```

```
##   0.0077123436  5.859914  0.006607028  4.814049
##   0.0078551648  5.848068  0.006522310  4.815679
##   0.0079979860  5.851398  0.006460712  4.815679
##   0.0081408072  5.851398  0.006460712  4.815679
##   0.0082836284  5.851398  0.006460712  4.815679
##   0.0084264495  5.849998  0.006405500  4.823944
##   0.0085692707  5.869799  0.004452303  4.854518
##   0.0087120919  5.870708  0.004469234  4.856787
##   0.0088549131  5.870708  0.004469234  4.856787
##   0.0089977343  5.870708  0.004469234  4.856787
##   0.0091405554  5.868524  0.004279589  4.852577
##   0.0092833766  5.867200  0.003767682  4.855436
##   0.0094261978  5.867200  0.003767682  4.855436
##   0.0095690190  5.850360  0.004609678  4.853306
##   0.0097118401  5.850360  0.004609678  4.853306
##   0.0098546613  5.849583  0.004607040  4.853777
##   0.0099974825  5.827574  0.004409660  4.832295
##   0.0101403037  5.827574  0.004409660  4.832295
##   0.0102831249  5.808543  0.004063088  4.806258
##   0.0104259460  5.803250  0.004010432  4.807494
##   0.0105687672  5.803250  0.004010432  4.807494
##   0.0107115884  5.796839  0.005141344  4.802801
##   0.0108544096  5.761189  0.006477761  4.785133
##   0.0109972308  5.761189  0.006477761  4.785133
##   0.0111400519  5.739434  0.006085092  4.774091
##   0.0112828731  5.739434  0.006085092  4.774091
##   0.0114256943  5.727860  0.006552259  4.770599
##   0.0115685155  5.735686  0.007444615  4.779745
##   0.0117113366  5.735686  0.007444615  4.779745
##   0.0118541578  5.735686  0.007444615  4.779745
##   0.0119969790  5.735686  0.007444615  4.779745
##   0.0121398002  5.735686  0.007444615  4.779745
##   0.0122826214  5.735686  0.007444615  4.779745
##   0.0124254425  5.737380  0.007721122  4.779760
##   0.0125682637  5.618228  0.010666536  4.667854
##   0.0127110849  5.618228  0.010666536  4.667854
##   0.0128539061  5.622070  0.009573647  4.671405
##   0.0129967273  5.622070  0.009573647  4.671405
##   0.0131395484  5.599498  0.010457787  4.649342
##   0.0132823696  5.599498  0.010457787  4.649342
##   0.0134251908  5.586050  0.010454727  4.651394
##   0.0135680120  5.586050  0.010454727  4.651394
##   0.0137108331  5.602147  0.008420945  4.668261
##   0.0138536543  5.602147  0.008420945  4.668261
##   0.0139964755  5.575695  0.010524161  4.655115
##   0.0141392967  5.584684  0.009003187  4.662808
##
## RMSE was used to select the optimal model using the smallest value.
## The final value used for the model was cp = 0.01399648.
```

---

9. Hyperparameter Explanation (Cost Complexity – cp)

The key hyperparameter in CART is **cost complexity (cp)**.

- Controls tree depth and pruning
- Penalizes excessive model complexity
- Balances bias vs variance

  A lower `cp` allows deeper trees, while a higher `cp` enforces simpler models.

The optimal `cp` is selected based on **minimum cross-validated RMSE**.

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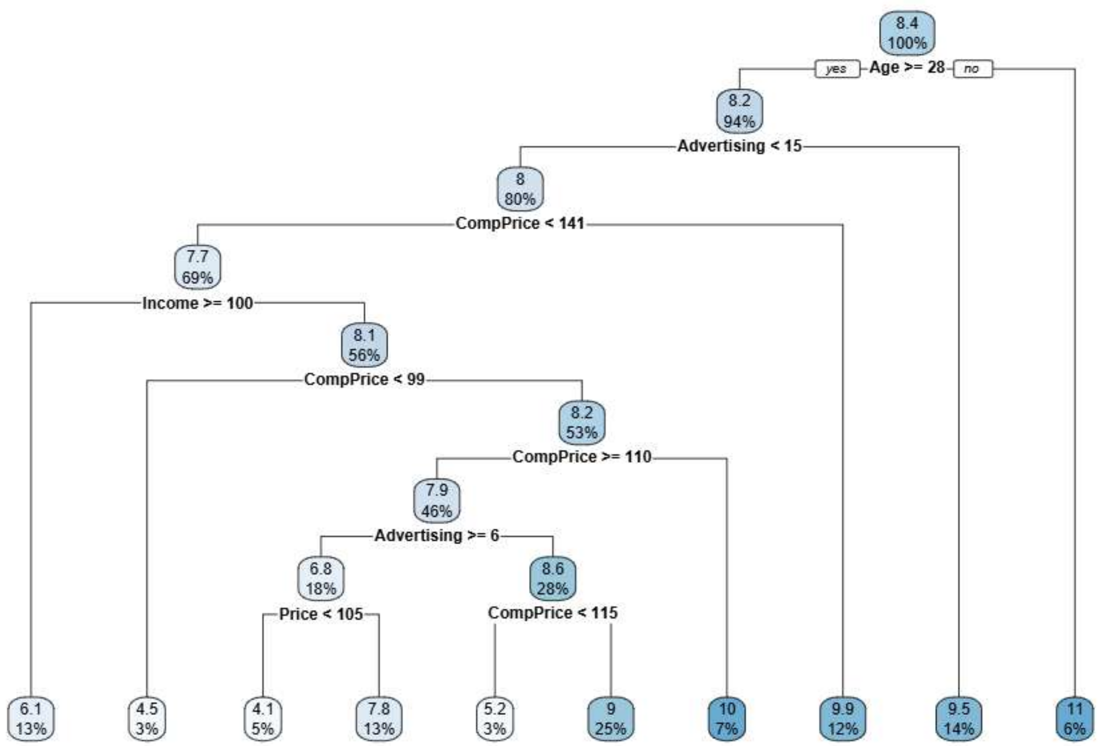
10. Model Visualization

```r
library(rpart.plot)

rpart.plot(cart_model$finalModel)
```



The CART regression tree reveals that **Shelf Location**, **Price**, **Advertising**, and **Competitor Pricing** are the most influential variables affecting car seat sales.

Early splits emphasize merchandising and pricing decisions, indicating that in-store placement and competitive pricing strategy play a dominant role in sales performance.

---

11. Model Performance Summary

```r
cart_model$results %>%
  arrange(RMSE) %>%
  head()
```

```
##            cp     RMSE   Rsquared      MAE    RMSESD  RsquaredSD      MAESD
```

```
## 1 0.01399648 5.575695 0.010524161 4.655115 0.4068420 0.006196913 0.2927815
## 2 0.01413930 5.584684 0.009003187 4.662808 0.3999887 0.005551365 0.2888911
## 3 0.01342519 5.586050 0.010454727 4.651394 0.3982953 0.006093235 0.2976736
## 4 0.01356801 5.586050 0.010454727 4.651394 0.3982953 0.006093235 0.2976736
## 5 0.01313955 5.599498 0.010457787 4.649342 0.4181625 0.006088946 0.2946992
## 6 0.01328237 5.599498 0.010457787 4.649342 0.4181625 0.006088946 0.2946992
```

- **RMSE** – prediction error magnitude
- **R-squared** – explained variance

---

12. Generating Predictions on New Store Data

In real-world retail environments, forecasting models are often applied to **new or upcoming stores** where historical sales data is unavailable.

To simulate this scenario, the trained CART model is applied to a new dataset containing store attributes without observed sales values. This enables proactive demand estimation prior to store launch or product rollout.

```r
# Load new, unseen data
new_car_seat_sales <- read.csv("data/new_car_seat_sales_data.csv")

# Inspect the structure
str(new_car_seat_sales)
```

```
## 'data.frame':    3 obs. of  11 variables:
##  $ Store      : int  216 8 273
##  $ CompPrice  : int  116 136 113
##  $ Income     : int  83 81 33
##  $ Advertising: int  15 15 0
##  $ Population : int  170 425 14
##  $ Price      : int  144 120 63
##  $ ShelfLoc   : chr  "Bad" "Good" "Good"
##  $ Age        : int  71 67 38
##  $ Education  : int  11 10 12
##  $ Urban      : chr  "Yes" "Yes" "Yes"
##  $ US         : chr  "Yes" "Yes" "No"
```

```r
new_model_data <- new_car_seat_sales %>%
select(-Store)
```

Predict sales for new stores

```r
new_predictions <- predict(cart_model, new_model_data)
```

Combine predictions with input data

```r
prediction_results <- new_car_seat_sales %>%
mutate(Predicted_Sales = new_predictions)

head(prediction_results)
```

```
##   Store CompPrice Income Advertising Population Price ShelfLoc Age Education
## 1   216       116     83          15        170   144      Bad  71        11
## 2     8       136     81          15        425   120     Good  67        10
## 3   273       113     33           0         14    63     Good  38        12
##   Urban  US Predicted_Sales
## 1   Yes Yes        9.527273
## 2   Yes Yes        9.527273
## 3   Yes  No        5.250000
```

Predicted_Sales represents the expected number of car seats sold (measured in **hundreds of units**) for each store configuration.

---

13. Business Insights

By applying the CART model to new store-level data, decision-makers can proactively estimate expected car seat demand before product deployment.

This enables:

- Pre-launch inventory optimization
- Location-specific pricing adjustments
- Targeted advertising allocation
- Risk reduction for underperforming store profiles

The interpretability of CART allows business stakeholders to directly understand **why** certain stores are forecasted to perform better than others.